MANASI MAHENDRA JESTE

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INTRODUCTION:

Master's student in Embedded Systems Engineering with a strong foundation in automation, embedded systems, and intelligent data analysis. Combines hands-on experience in programming, sensors, and control-oriented systems with research in machine learning, forecasting, and optimization for EV charging and energy-related applications. Motivated to contribute to roles in automation, renewable energy, sustainability, AI, and data science.

EDUCATION:

M.Sc. in Embedded Systems Engineering
Albert-Ludwigs-Universität Freiburg | 10/2023 – Present
Specialization: AI | Grade: 2.4

B.Tech. in Instrumentation and Control Engineering
Vishwakarma Institute of Technology, Pune | 08/2018 – 07/2022
Grade: 1.2

PROJECTS:

- **EcoAutoVision**: CO_2-aware AutoML for vision models Resource-aware AutoML with proxy-based candidate ranking, multi-objective HPO, and dynamic training policies to reduce GPU hours and emissions.
- **Smart Stretcher**: IES Lab Freiburg R&D and UI development for contactless, non-invasive patient monitoring in ambulances.
- **Two-Point Security System for Doors/Lockers:** Machine learning and IoT-based security system using Python and OpenCV.
- **Remote Bootloading System for NXP Controllers:** Developed a remote bootloading system in C for Battery Management System for Electric Vehicle.
- **Image Telemetry System:** Developed Python algorithms for image compression, data splitting, and transmission with real-time telemetry integration.
- **Building Automation Projects:** Software development and simulation for HVAC systems including AHUs, chillers, and FCUs for Sydney Airport Terminal 3 and Grand Hyatt Singapore.
- **Robotics and Control Projects:** Included PID tuning, robotic arm programming, DC motor position control, pantograph and slider-crank simulation, omni/mechanum chassis locomotion, and SimScape robotic arm modeling.
- **Motorized Autofocus for Surveillance:** Image-processing and control-based autofocus system for surveillance applications.
- **Integrated Health Monitoring System** Python and machine learning-based health monitoring project

INTERESTS

- Automation & Control
- Renewable Energy
- Sustainability
- EV Charging Systems
- Machine Learning & AI
- Data Science & Forecasting

SKILLS

- Python, C, C++
- MATLAB / Simulink
- Machine Learning
- Forecasting & Optimization
- Embedded Systems
- Sensors & Microcontrollers
- Data Analysis
- HVAC / Control Systems

STRENGTHS

- Problem Solving
- Analytical Thinking
- Research
- Communication
- Teamwork
- Adaptability

LANGUAGE

- English C1
- German B1.1
- Hindi
- Marathi

EXPERIENCE:

1. **Student Research Assistant:**
   INATECH, University of Freiburg, Germany | 10/2025– Present
   - Designed and tuned PID controllers for heat pump systems for different operating configurations
   - Developed modeling logic, simulation, and testing workflows for monovalent and bivalent heat pump systems.
   - Worked on validation and performance analysis for small and large-scale heat pumps under different system scenarios.
2. **Master Thesis Student:**
   Fraunhofer ISE, Freiburg, Germany | 10/2025 – Present
   - Developed AI-based probabilistic forecasting models for EV charging sessions, focusing on departure time, dwell time, and energy demand for smart charging applications.
   - Built and evaluated Diffusion, Transformer, and LSTM-based time series models using real-world EV charging datasets, with emphasis on uncertainty calibration and operational performance.
   - Translated forecasting results into practical charging-station KPIs such as occupancy estimation and risk-aware energy planning for smarter station operation.
3. **Student Research Assistant**:
   Fraunhofer ISE, Freiburg, Germany | 01/2024 – 10/2025
   - Researched V2X systems and smart EV charging optimization
   - Built forecasting models for EV arrival/departure prediction using real-world datasets
   - Applied MILP and IPOPT for charging schedule optimization
4. **Systems Engineer**
   Honeywell Automation India Ltd, Pune, India | 07/2022 – 09/2023
   - Programmed and simulated HVAC control systems for large-scale infrastructure
   - Delivered automation solutions for AHUs, chillers, and FCUs
   - Worked on projects including Sydney Airport and Grand Hyatt Singapore
5. **Research Intern**
   Automotive Research Association of India (ARAI), Pune, India | 01/2022 – 07/2022
   - Contributed to battery management system development
   - Developed a remote bootloading system for NXP controllers
6. **Research Intern**
   Defence Research and Development Organization (DRDO), Pune, India | 07/2021 – 12/2021
   - Developed Python algorithms for image compression, splitting, and transmission
   - Supported real-time image telemetry integration with base-station systems

ACHIEVEMENTS:

- Recognition in Badische Zeitung for the Smart Stretcher project at IES Lab Freiburg.
- Participation in COSIMA national competition with the Smart Stretcher project.
- ISA Educational Scholarship Award 2021 for excellence in automation and instrumentation.
- Best Research Paper Award for *Effects of sodium selenite on growth of Pleurotus Sajor-Caju* at a research conference in Panjim, Goa.
- Team Lead, ROBOCON 2020 and 2021, focused on MATLAB/Simulink simulation and multi-sensor robotics integration.

PUBLICATIONS:

- Mathematical Modeling and Simulation of Pantograph Mechanism in MATLAB
  Manasi Jeste, Aniruddha Atre, *International Journal of Engineering Research & Technology (IJERT)*, Vol. 10, Issue 07, July 2021. Paper ID: IJERTV10IS070284.
- Two-Point Security System for Doors/Lockers Using Machine Learning and IoT
  Presented at an IEEE conference in Coimbatore.
- Effects of Sodium Selenite on Growth of Pleurotus Sajor-Caju
  Presented at a conference in Panjim, Goa and published by A.R. Research Publication / International Journal of Management, Technology and Engineering, Volume 8, Issue X, October 2018.